Lecture – 8
EE10510

Single-Phase Transformer

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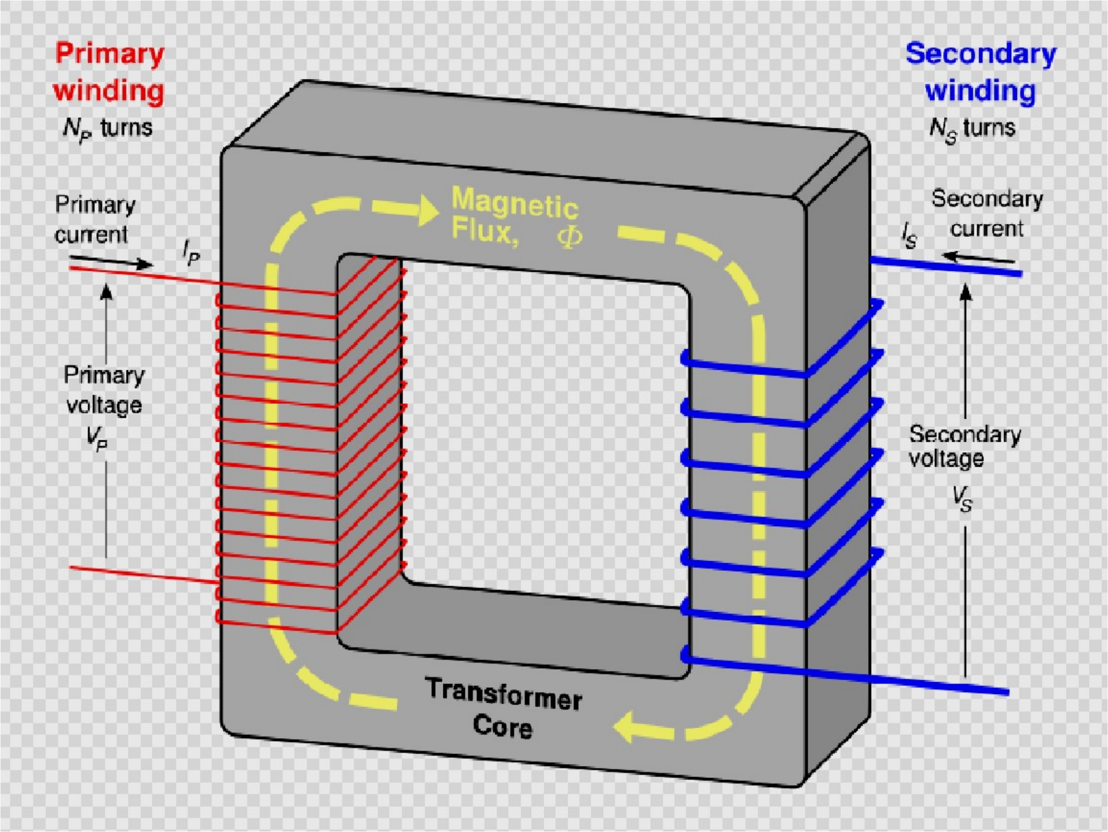

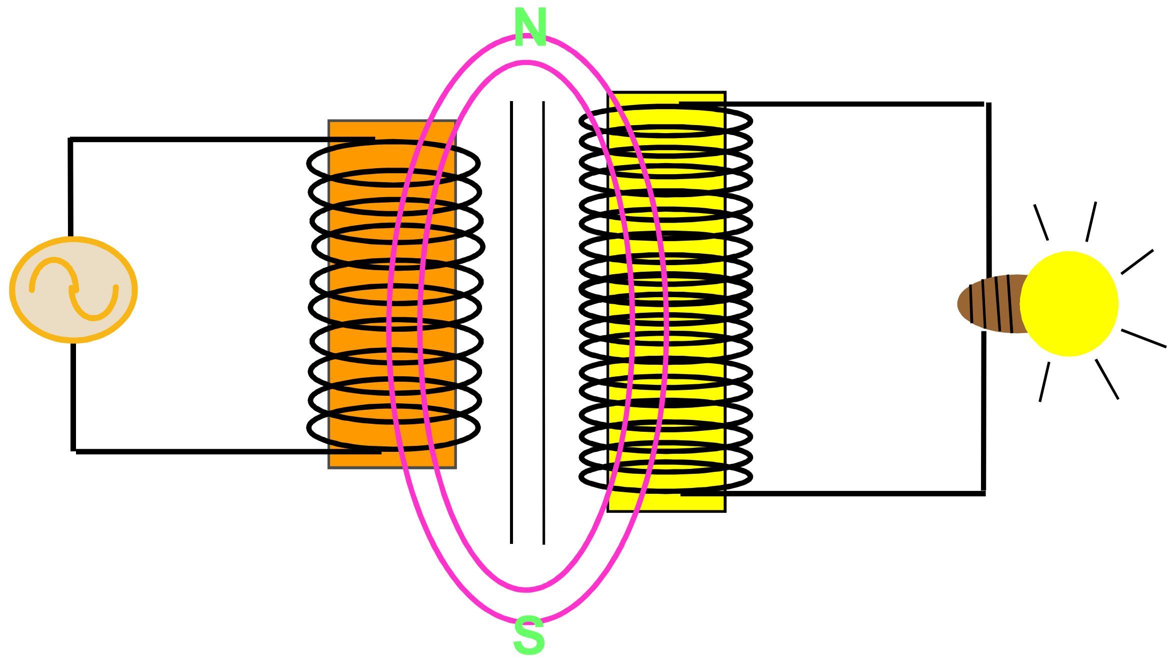

A transformer can change electrical energy of a given voltage into electrical energy at a different voltage level. It consists of two coils arranged in such a way that the magnetic field surrounding one coil cuts through the other coil. When an **alternating voltage** is applied to (across) one coil, the **varying magnetic field** set up around that coil induces an alternating voltage in the other coil.

- Transformers will not work with direct current, since no changing magnetic field is produced, and therefore no current can be induced.
- The factor which determines whether a transformer is a step up (increasing the voltage) or step down (decreasing the voltage) type decide the "turns" ratio.
- The turns ratio is the ratio of the number of turns in the primary winding to the number of turns in the secondary winding.

Ideal Transformer

In order to have better understanding of the behavior of the transformer, initially certain idealizations are made and the resulting 'ideal' transformer is studied. These idealizations are as follows:

- Magnetic circuit is linear and has infinite permeability. The consequence is that a vanishingly small current is enough to establish the given flux. Hysteresis loss is negligible. As all the flux generated confines itself to the iron, there is no leakage flux.
- Windings do not have resistance. This means that there are no copper losses, nor there is any ohmic drop in the electric circuit.

Ideal Single-Phase Transformer (No-Load Operation)

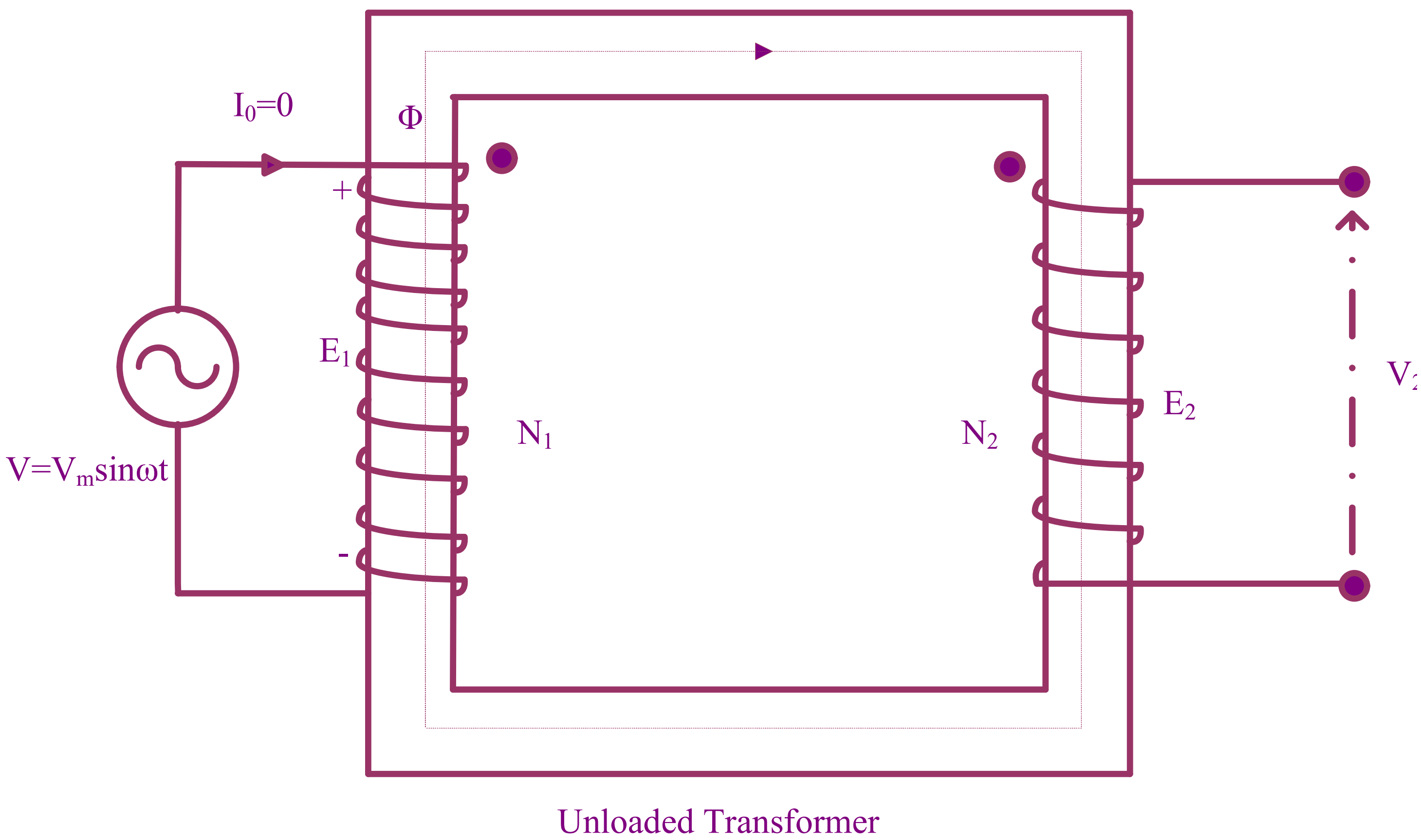

Fig. 1

- Fig. 1 shows a two winding ideal transformer. The primary winding has N_1 turns and is connected to a voltage source of V_1 volts.
- The secondary has N_2 turns. Secondary can be connected to a load impedance for loading the transformer.
- The primary and secondary are shown on the different limb and separately for clarity.
- As a current I_0 amps is passed through the primary winding of N_1 turns it sets up an mmf of I_0N_1 ampere which is in turn sets up a flux through the core.

- Since the reluctance of the iron path given by R = l/μ A is zero as μ is infinity, a vanishingly small value of current I_0 is enough to setup a flux which is finite.
- As I_0 establishes the field inside the transformer, it is called the magnetizing current of the transformer.

$$\text{Flux } \phi = \frac{\text{mmf}}{\text{reluctance}} = \frac{I_0 N_1}{l / \mu A} = \frac{I_0 N_1 \mu A}{l}$$

- This current is the result of a sinusoidal voltage V applied to the primary.

- As the current through the loop is zero (or vanishingly small), at every instant of time, the sum of the voltages must be zero inside the same. Writing this in terms of instantaneous values we have,

$$v_1 - e_1 = 0$$

- where v_1 is the instantaneous value of the applied voltage and e_1 is the induced emf due to Faradays principle. The negative sign is due to the application of the Lenz's law and shows that it is in the form of a voltage drop. Kirchhoff's law application to the loop will result in the same thing.

- This equation results in $E_1 = E_1$ or the induced emf must be same in magnitude to the applied voltage at every instant of time.
- Let $e_1 = V_m \sin\omega t$ where V_m is the peak value and $\omega = 2\pi f$ t. where f is the frequency of the supply. As $v_1 = e_1$; $e_1 = d\,\Phi/dt$
- but $e_1 = E_m \sin\omega t$, $e_1 = v_1$. It can be easily seen that the variation of flux linkages can be obtained as $\Phi = \Phi_m \cos\omega t$. Here Φ_m is the peak value of the flux linkages of the primary.
- Thus the RMS primary induced emf is $$e_1 = \frac{d\phi}{dt} = \frac{d\phi_m \cos\omega t}{dt}$$ $$e_1 = -\phi_m \omega \sin\omega t$$

The rms value of the primary induced voltage is

$$\phi_m \omega / \sqrt{2} = 2\pi f N_1 \phi_m / \sqrt{2} = 4.44 f N_1 \phi_m \text{ volts}$$

Similarly rms value of secondary side induced voltage is given by

$$\phi_m \omega / \sqrt{2} = 2\pi f N_2 \phi_m / \sqrt{2} = 4.44 f N_2 \phi_m \text{ volts}$$

where $\mathrm{B}_{\max} = \phi_{\max} / A$

Which yields the voltage ratio as,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

- The voltages E_1 and E_2 are obtained by the same mutual flux and hence they are in phase.
- If the winding sense is opposite i.e., if the primary is wound in clockwise sense and the secondary counter clockwise sense then if the top terminal of the first winding is at maximum potential the bottom terminal of the second winding would be at the peak potential.

Ex.1 A single phase transformer is connected to a 660 V supply. The voltage/turn of the transformer is 1.1V. The secondary voltage of the transformer is found to be 440V . Determine the primary and secondary turns and area of cross section of the core if the maximum flux density is 1.4T.

Sol.

$$N_1=(E_1/1.1)=(660/1.1)=600$$

$$N_2=(E_2/1.1)=(440/1.1)=400$$

Again $E1=4.44fN_1B_{max}A$

$$\text{so } A=(E_1/4.44fN_1B_{max})=(660/4.44*50*600*1.4)$$

$$=35.39\text{cm}^2$$

References

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